```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
a=np.array([1,2,3])
a
```

```
array([1, 2, 3])
```

```
a=np.array([[1,2,3],[4,5,6]])
a
```

```
array([[1, 2, 3],
       [4, 5, 6]])
```

```
a=np.full(2,4)
a
```

```
array([4, 4])
```

```
a=np.full((2,2),7)
a
```

```
array([[7, 7],
       [7, 7]])
```

```
a=np.eye(3)
a
```

```
array([[1., 0., 0.],
       [0., 1., 0.],
       [0., 0., 1.]])
```

```
a=np.zeros(2)
a
```

```
array([0., 0.])
```

```
a=np.zeros((2,2))
a
```

```
array([[0., 0.],
       [0., 0.]])
```

```
a=np.ones(3)
a
```

```
array([1., 1., 1.])
```

```
a=np.ones((2,4))
a
```

```
array([[1., 1., 1., 1.],
       [1., 1., 1., 1.]])
```

```
a=np.array([1,10,78])
a
```

```
array([ 1, 10, 78])
```

```
b=np.diag(a)
b
```

```
array([[ 1,  0,  0],
       [ 0, 10,  0],
       [ 0,  0, 78]])
```

```
a=np.array([1,2,3,1,1,3,3,4,2,3,4,5])
```

```
np.unique(a,return_index=True,return_counts=True)
```

```
(array([1, 2, 3, 4, 5]), array([ 0,  1,  2,  7, 11]), array([3, 2, 4, 2,
1]))
```

```
a=np.arange(1,10).reshape(3,3)
a
```

```
array([[1, 2, 3],
       [4, 5, 6],
       [7, 8, 9]])
```

```
print(np.sum(a))
print(a.sum(axis=1))
print(np.sum(a,axis=0))
```

```
45
[ 6 15 24]
[12 15 18]
```

```
a=np.random.randint(1,10,3)
a
```

```
array([3, 8, 2])
```

```
np.random.seed(1)
a=np.random.randint(1,10,3)
a
```

```
array([6, 9, 6])
```

```
a=np.random.rand(3)
a
```

```
array([0.34556073, 0.39676747, 0.53881673])
```

```
a=np.arange(1,19,2)
a
```

```
array([ 1,  3,  5,  7,  9, 11, 13, 15, 17])
```

```
a=np.array([1,2,3,4,5,6,7,8,9,10,11,12])
a
```

```
array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12])
```

```
b=np.reshape(a,(2,6))
b
```

```
array([[ 1,  2,  3,  4,  5,  6],
       [ 7,  8,  9, 10, 11, 12]])
```

```
a=np.array([[1,2,3],[4,5,6]])
np.transpose(a)
```

```
array([[1, 4],
       [2, 5],
       [3, 6]])
```

```
a=np.random.randint(1,100,size=(2,3))
a
```

```
array([[19, 85, 12],
       [29, 30, 15]])
```

```
a=np.arange(1,19).reshape(6,3)
a
```

```
array([[ 1,  2,  3],
       [ 4,  5,  6],
       [ 7,  8,  9],
       [10, 11, 12],
       [13, 14, 15],
       [16, 17, 18]])
```

```
#pandas
```

```
df=pd.read_csv('/content/covid_toy.csv')
df.head()
```

	age	gender	fever	cough	city	has_covid
0	60	Male	103.0	Mild	Kolkata	No
1	27	Male	100.0	Mild	Delhi	Yes
2	42	Male	101.0	Mild	Delhi	No
3	31	Female	98.0	Mild	Kolkata	No
4	65	Female	101.0	Mild	Mumbai	No

Next steps: Generate code with df | New interactive sheet

```
df.tail()
```

	age	gender	fever	cough	city	has_covid
95	12	Female	104.0	Mild	Bangalore	No
96	51	Female	101.0	Strong	Kolkata	Yes
97	20	Female	101.0	Mild	Bangalore	No
98	5	Female	98.0	Strong	Mumbai	No
99	10	Female	98.0	Strong	Kolkata	Yes

```
df.sample(5)
```

	age	gender	fever	cough	city	has_covid
74	34	Female	104.0	Strong	Delhi	No
45	72	Male	99.0	Mild	Bangalore	No
48	66	Male	99.0	Strong	Bangalore	No
31	83	Male	103.0	Mild	Kolkata	No
65	69	Female	102.0	Mild	Bangalore	No

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 6 columns):
 #   Column     Non-Null Count  Dtype  
---  ------     --------------  -----  
 0   age        100 non-null    int64  
 1   gender     100 non-null    object 
 2   fever      90 non-null     float64
 3   cough      100 non-null    object 
 4   city       100 non-null    object 
 5   has_covid  100 non-null    object 
```

```
dtypes: float64(1), int64(1), object(4)
memory usage: 4.8+ KB
```

```
df.columns
```

```
Index(['age', 'gender', 'fever', 'cough', 'city', 'has_covid'],
dtype='object')
```

```
df.dtypes
```

	0
age	int64
gender	object
fever	float64
cough	object
city	object
has_covid	object

dtype: object

```
df_ob=df.select_dtypes(include='object')
df_ob.columns.to_list()
```

```
['gender', 'cough', 'city', 'has_covid']
```

```
df_in=df.select_dtypes(include='int64')
df_in.columns.to_list()
```

```
['age']
```

```
df['cough'].value_counts()
```

cough	count
Mild	62
Strong	38

dtype: int64

```
df.isnull().sum()
```

	0
age	0
gender	0
fever	10
cough	0
city	0
has_covid	0

dtype: int64

```
df.describe()
```

	age	fever
count	100.000000	90.000000
mean	44.220000	100.844444
std	24.878931	2.054926
min	5.000000	98.000000
25%	20.000000	99.000000
50%	45.000000	101.000000
75%	66.500000	102.750000
max	84.000000	104.000000

```
df.shape
```

```
(100, 6)
```

```
df.size
```

```
600
```

```
df.filter(items=['fever','age'])
```

	fever	age
0	103.0	60
1	100.0	27
2	101.0	42
3	98.0	31
4	101.0	65
...	...	...
95	104.0	12
96	101.0	51
97	101.0	20
98	98.0	5
99	98.0	10

100 rows × 2 columns

```
df[['age','fever']]
```

	age	fever
0	60	103.0
1	27	100.0
2	42	101.0
3	31	98.0
4	65	101.0
...	...	...
95	12	104.0
96	51	101.0
97	20	101.0
98	5	98.0
99	10	98.0

100 rows × 2 columns

```
df_loc=df.loc[:4,['fever']]
df_loc
```

	fever
0	103.0
1	100.0
2	101.0
3	98.0
4	101.0

```
df_iloc=df.iloc[:4,:4]
df_iloc
```

	age	gender	fever	cough
0	60	Male	103.0	Mild
1	27	Male	100.0	Mild
2	42	Male	101.0	Mild
3	31	Female	98.0	Mild

```
df['cough'].value_counts()
```

cough	count
Mild	62
Strong	38

dtype: int64

```
df['cough']=df['cough'].map({'Mild':0,'Strong':1})
```

```
df.head(7)
```

```
df.rename(columns={'fever':'update_fever'})
```

	age	gender	update_fever	cough	city	has_covid
1	27	Male	100.0	0	Delhi	Yes
0	60	Male	103.0	0	Kolkata	No
1	27	Male	100.0	0	Delhi	No
3	31	Female	98.0	0	Kolkata	Yes
2	42	Male	101.0	0	Delhi	No
3	31	Female	98.0	0	Kolkata	No
5	84	Female	NaN	0	Bangalore	No
4	65	Female	101.0	0	Mumbai	No
...	...	...	...	...	...	...
95	12	Female	104.0	0	Bangalore	No
96	51	Female	101.0	1	Kolkata	Yes
97	20	Female	101.0	0	Bangalore	No
98	5	Female	98.0	1	Mumbai	No
99	10	Female	98.0	1	Kolkata	Yes

100 rows × 6 columns

```
df['has_covid_new']=df['has_covid'].where(df['has_covid']=='Yes',other='negat
```

```
df.head(10)
```

	age	gender	fever	cough	city	has_covid	has_covid_new
0	60	Male	103.0	Mild	Kolkata	No	negative
1	27	Male	100.0	Mild	Delhi	Yes	Yes
2	42	Male	101.0	Mild	Delhi	No	negative
3	31	Female	98.0	Mild	Kolkata	No	negative
4	65	Female	101.0	Mild	Mumbai	No	negative
5	84	Female	NaN	Mild	Bangalore	Yes	Yes
6	14	Male	101.0	Strong	Bangalore	No	negative
7	20	Female	NaN	Strong	Mumbai	Yes	Yes
8	19	Female	100.0	Strong	Bangalore	No	negative